

Question 1

$$a) \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix}$$

b) coefficient matrix variable matrix constant matrix

$$c) \left[\begin{array}{ccc|c} 2 & -1 & 3 & 5 \\ 1 & 2 & -1 & 4 \\ 3 & 1 & 2 & 1 \end{array} \right]$$

Question 2

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 2 & 2 \end{array} \right] \begin{array}{l} \sim R_2 - 2R_1 \\ \sim R_3 + R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 \end{array} \right] \sim R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 2 \end{array} \right] \sim R_2 - R_3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{array} \right] \sim R_2 - R_3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right] \sim R_3 \times \frac{1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right] \sim R_{23}$$

Question 3

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array} \right] \sim R_2 - 2R_1, R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & -1 & -9 \end{array} \right] \sim R_{23}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array} \right] \sim R_3 + 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$z = 3$$

$$y - 2(3) = -4$$

$$y = 2$$

$$x + 3 + 2 = 6$$

$$x = 1$$

$$(1, 2, 3)$$

Question 4

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 1 & -1 & 2 & 4 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & -3 & 3 & 1 \end{array} \right] \sim R_2 - 2R_1, R_3 - R_1$$

hence, there are infinitely many solutions.

$$1 + 3y = 3t$$

$$z = t$$

$$y = \frac{3t - 1}{3}, x = \frac{11 - 3t}{3}$$

$$-3y + 3z = 1 \rightarrow y = \frac{3z - 1}{3}$$

$$x + 2y - z = 3$$

$$x + 2\left(\frac{3z - 1}{3}\right) - z = 3$$

$$x + 2\left(z - \frac{1}{3}\right) - z = 3$$

$$x + z - \frac{2}{3} = 3$$

$$x + z = \frac{11}{3}$$

$$3x + 3z = 11$$

$$3x = 11 - 3z$$

Question 5

$$a) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \sim R_2 - 3R_1$$

$$b) E, A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+0 & 2+0 \\ -3+3 & -6+4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} \sim R_2 - 3R_1$$

$$c) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \sim R_1 + 2R_2$$

Question 6

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & -1 & 1 & 0 & 1 & 0 \\ 1 & 2 & -1 & 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -3 & -1 & -2 & 1 & 0 \\ 0 & 1 & -2 & -1 & 0 & 1 \end{array} \right] \sim \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 2 & 0 & -1 \\ 0 & 0 & -7 & -5 & 1 & 3 \\ 0 & 1 & -2 & -1 & 0 & 1 \end{array} \right] \sim \begin{array}{l} R_1 - R_3 \\ R_2 + 3R_3 \end{array} = \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 2 & 0 & -1 \\ 0 & 1 & -2 & -5 & 1 & 3 \\ 0 & 0 & -7 & -5 & 1 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 2 & 0 & -1 \\ 0 & 1 & -2 & -5 & 1 & 3 \\ 0 & 0 & 1 & 5/7 & -1/7 & -3/7 \end{array} \right] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/7 & 3/7 & 2/7 \\ 0 & 1 & 0 & 3/7 & -2/7 & 1/7 \\ 0 & 0 & 1 & 5/7 & -1/7 & -3/7 \end{array} \right] \sim \begin{array}{l} R_1 - 3R_3 \\ R_2 + 2R_3 \end{array}$$

Question 7

- a) A: Diagonal, Triangular, Symmetric
B: Upper Triangular
C: Symmetric

b) $|A| = 2 \times (-3) \times 4 = -24$

$|B| = 1 \times 4 \times 2 = 8$

$|C| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & -1 \\ 0 & -1 & -3 \end{vmatrix} \begin{matrix} \sim R_2 - 2R_1 \\ \sim R_3 - 3R_1 \end{matrix}$

$\begin{vmatrix} 0 & -1 \\ -1 & 3 \end{vmatrix} = 0 - (-1)(-1) = -1$

All matrices are invertible

c) $A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & -1/5 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}$

- d) None of the diagonal entries are zero.

Question 8

$$a) \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

$$b) \begin{bmatrix} 2(1) - 2 \\ 1 + 3(2) \end{bmatrix} = \begin{bmatrix} 0 \\ 7 \end{bmatrix}$$

$$c) T(1,0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad T(0,1) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$d) 2(3) - (-1)(1) = 7$$

The determinant is non zero so it is invertible

Question 9

a) $(0, 1): c = 4$

$(1, 4): a + b + c = 4$

$(2, 9): 4a + 2b + c = 9$

b)
$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}$$

c)
$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 9 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 2 & 4 & 8 \end{array} \right] \sim R_2 - R_1, R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \end{array} \right] \sim R_3 - 2R_2$$

$$a = 1$$

$$b + 1 = 3$$

$$c = 1$$

$$\therefore b = 2$$

$$a = 1, b = 2, c = 1$$

d) $p(3) = 3^2 + 2(3) + 1 = 9 + 6 + 1 = 16$

Question 10

$$a) \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 19 \\ 20 \end{bmatrix}$$

$$b) \begin{bmatrix} 1 & 2 & 1 & | & 12 \\ 2 & 4 & 3 & | & 19 \\ 3 & 2 & 2 & | & 20 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & | & 12 \\ 0 & 0 & 1 & | & -5 \\ 0 & -4 & -1 & | & -16 \end{bmatrix} \begin{matrix} \\ \sim R_2 - 3R_1 \\ \sim R_3 - 3R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & | & 12 \\ 0 & -4 & -1 & | & -16 \\ 0 & 0 & 1 & | & -5 \end{bmatrix} \sim R_{23} \Rightarrow z = -5$$

$$-4y + 5 = -16 \therefore y = 21/4$$

$$x + \frac{21}{4} - 5 = 12 \therefore x = 6.5$$

$$x = 6.5, y = 5.25, z = -5$$

$$c) 6.5 + 2\left(\frac{21}{4}\right) - 5 = 12 \checkmark$$

$$2(6.5) + 4\left(\frac{21}{4}\right) + 3(-5) = 19 \checkmark$$

$$3(6.5) + 2\left(\frac{21}{4}\right) + 2(-5) = 20 \checkmark$$

d) Because the relationship between variables in A remains unchanged

Question 12

$$a = 7$$

$$b = 0$$

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 8x - 0y \\ 7x + y \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 7 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$b) \begin{bmatrix} 8(1) - 0 \\ 7(1) + 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 9 \end{bmatrix}$$

$$c) |A| = 8 - 0 = 8 ; \text{ It is invertible}$$

$$d) \left[\begin{array}{cc|cc} 8 & 0 & 1 & 0 \\ 7 & 1 & 0 & 1 \end{array} \right] = \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{8} & 0 \\ 7 & 1 & 0 & 1 \end{array} \right] \sim R_1 \times \frac{1}{8}$$

$$= \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{8} & 0 \\ 0 & 1 & \frac{7}{8} & 1 \end{array} \right] \sim R_2 + 7R_1$$

Question 11

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 10 \\ 2 & 1 & 1 & 8 \\ 1 & -1 & 2 & 5 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 10 \\ 0 & -3 & -1 & -12 \\ 0 & -3 & 1 & -5 \end{array} \right] \begin{array}{l} \sim R_2 - 2R_1 \\ \sim R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 10 \\ 0 & -3 & -1 & -12 \\ 0 & 0 & 2 & 7 \end{array} \right] \sim R_3 - R_2$$

$$2z = 7$$

$$z = \frac{7}{2}$$

$$-3y - \frac{7}{2} = -12$$

$$-3y = -9 + \frac{1}{2}$$

$$y = 3 - \frac{1}{6}$$

$$y = \frac{18-1}{6} = \frac{17}{6}$$

$$x + 2\left(\frac{17}{6}\right) + \frac{7}{2} = 10$$